Exam questions

Question 1: Cap-rates and WACC can be useful short-cut methods to determine the viability of a real estate investment.

a- Explain how cap-rates can be applied to determine the *purchase* and *sale* price of an investment when you have information about the Net Operating Income and cap-rates of comparable properties in a comparable market. (10 points)

These issues were extensively discussed in Lecture 2, including several examples taken from the books in the reading list.

b- How do cap-rates vary to reflect risks and opportunities of a real estate investment? And in what sense does a cap-rate condense the calculations you would carry out in a fully-fledged discounted cash flow (DCF) approach into one single operation? With what advantages and risks? (15 points)

These issues were also extensively discussed in Lecture 2.

c- Focus on the office sector and discuss how the structure of buildings, features of the leases and characteristics of tenants, and the location of buildings affect cap-rates. (15 points)

These issues were also extensively discussed in Lecture 2.

d- Consider the following investment. Its price is £200 million and the unadjusted cap-rate is 3%. Nominal growth in rents is assumed to be negative, at -2% per year. You expect to hold the investment of 5 years and then sell it (in year 5). Determine the sale price and the IRR on this investment. (20 points)

The first calculations students have to do is to determine the initial NOI. This is found by multiplying the initial price (£200 million) by the corrected cap-rate (3%-(-2%)=5%. The initial NOI is thus £10 million. Then it decreases over time by 2%. The calculations need to be carried out till 6 years after the initial investment – that is one period after sale, to determine the relevant price and calculate the sale price. See tabulation below:

	T=0	T=1	T=2	T=3	T=4	T=5	T=6
NOI		10	9.80	9.60	9.41	9.22	9.04
Sale/Purchase price	-200					180.78	

The IRR on this investment can be obtained using the Cash Flow worksheet and it's 3%.

e- Compare the IRR with the unadjusted and the adjusted (Gordon model) cap rate. What do you notice? Are these results expected? (10 points)

The results are expected and driven by the fact that a Gordon-model approach corrects for the negative growth in the NOI by subtracting it from the cap-rate. Conversely, the DCF does that explicitly in the calculations of the NOI.

f- You now leverage your investment and take out an interest-only loan for 80% of the value of the investment and with an annual interest rate of 2% (compounding is on an annual basis). Using a WACC formula, calculate the cash-on-cash yield for the first and fifth year, and the annual growth yield on this investment. Comment on your findings. (20 points)

The calculations were explained during lectures 3 and 4. These are the findings:

Cash-on-cash year 1: (5%-80%*2%)/(1-80%)=17%

Cash-on-cash year 5: (4.61%-80%*2%)/(1-80%)=15.05%; note the cash yield in period 5 is given by 9.22/200=4.61%

Growth yield: over 5 years, this is equal to [(180.78-160)-(200-160)]/(200-160), where 160 is the amount borrowed (80% of 200). This is (20.78-40)/40 or -48.05% over five years. On an annualised basis this can be obtained as follows: (1+r)^5*40=20.78 → (20.78/40)^(1/5)-1 → -12.28%.

The students are expected to comment on the attractive cash yield and problematic negative growth yield. Good answers should also suggest that the overall IRR is a weighted average of all these figures and that the WACC is useful as an approximation but is affected by imprecision/myopia.

g- Suppose instead you leveraged at 95% of the value of the property (same interest rate). What would have happened to the equity in this investment at the time of sale? What would have been the possible implications of this finding? (10 points)

If the LTV was 95%, the amount borrowed would have been £190. Since the sale price is only 180.75, the borrower does not have enough to pay back the loan and is in negative equity. If lucky, he/she can extend the mortgage. Or else, he/she might have to default. This is the risk that these WACC calculations should flag and good students should reflect on this when answering this point.

Question 2: Direct investment in real estate has risks and opportunities. Investing in real estate *indirectly* allows you to reap some of the upsides while avoiding some of the downsides of real estate.

a- What are the main opportunities/upsides of directly investing in real estate? And what are instead the most important real-estate specific risks and downsides? (10 points)

These topics were discussed in lecture 1, 3 and then again in lecture 7 on REITS.

b- In what sense does a pro-forma analysis allow you to spell out all the components that determine the value of your investment, and the assumptions underpinning your assessment of the risks? (5 points)

These issues were discussed in lecture 2.

c- Rent roll, vacancies, other sources of income and operating expenses are the four items you consider in your pro-forma to determine the Net Operating Income (NOI) of your investment. Discuss each item in detail, and highlight how the four are interrelated. Enrich your answer with some of the considerations put forward by the 'case study' speakers. (20 points)

These issues were discussed in lecture 2, and then by both case study speakers. I expect students to talk about rent-free periods, overage rents, CPI indexed rents, allocation of communal spaces, as well as parking lots, laundry, advertisement as additional sources of income; plus vacancies affecting both rent but also operating expenses and additional sources of income.

d- Capital expenditures and capital improvements. Would you always consider these items as expenses or sometimes as investments? Discuss with the help of at least one example. (20 points)

These issues were discussed in lecture 2 (the specific examples given were the fitting out of the LSE bank and some of the redevelopment of the NAB). The two case study speakers touched on these issues too.

e- Which methods can you use to assess the risk of your real estate investment? (5 points)

These issues were discussed in Lecture 3 and several examples of scenario calculations were provided during the classes.

f- How does investing indirectly in real estate through REITs by-pass some of the risks that are specific to real estate? And what are the main constraints that REITs face in terms of management and investment? How can they 'add value' given these constraints? (20 points)

These were the core issues in the first part of Lecture 8 on REITs.

g- What are the main lessons in terms of viability and profitability of UK REITs drawn from their experience since the start in 2007? Discuss using the insights put forward during the intervention of the external lecturer. (20 points)

The second part of lecture 8 was delivered by Francis Salway, former CEO of Land Securities. He discussed the experience of UK REITs covering a number of aspects, including financial performance, leverage, profitability.

Question 4: You have decided to buy a property worth £1,000,000 and the bank has agreed to extend a loan for 70% of the value of the property (i.e. LTV=70%) over 30 years and with an annual interest rate of 8%. The loan has constant payments (CPM) and is fully amortizing. Initially you are offered the possibility of having either daily, monthly or quarterly compounding.

a. What would be the effective annual yield with quarterly, monthly and daily compounding (consider a year of 365 days)? (10 points)

These calculations were shown in Lecture 4. The numbers are as follows:

Monthly: (100) * [1+8%/12]^12 – 100 = 8.3%

Quarterly: (100) * [1+8%/4]^4 – 100 = 8.24%

Daily: (100) * [1+8%/365]^365 – 100 = 8.33%

b. What would be value of the first payment of the loan with the three different compounding (consider the 1st day payment for daily compounding, the 1st month payment with monthly compounding, and the 1st quarter payment with quarterly compounding)? (15 points)

These calculations follow those presented in Lectures 4 and 5. The payments would be as follows. Monthly: £5,136.35. Quarterly: £15,433.67. Daily: £168.74.

c. What would be the amounts paid in interest and in amortization at the end of year 1 for the loan with the three different compounding structures? And the outstanding amount of the loan at the end of year 1 with the three alternative compounding approaches? (10 points)

Similar calculations were presented in Lectures 4 and 5. The numbers are as follows.

Monthly: Int: £55,789; Amort: £5847; Bal: £694,152

Quarterly: Int: £55,825; Amort: £5909; Bal: £694,091

Daily: Int: £55,771; Amort: £5818; Bal: £694,182

d. You decide to go for the monthly compounding and pay in monthly instalments. However, you want to have the option to repay early some of the outstanding balance. In order to accommodate early repayments, the bank adds 3 discount points to the loan and charges you a 3% pre-payment fee for unscheduled principal reductions (the rest of the loan is the same: 30 year maturity; 70% LTV; 8% interest rate; monthly compounding and payments). What is the effective cost of borrowing if you hold the mortgage to maturity? And if you repay after 10 years? And after 5 years? (15 points)

These issues were covered in Lecture 5. To calculate the first figures, students need to consider that the PMTs stay at 5,136.35, but vis-à-vis a loan of 97% of 700,000 or £679,000. The IRR/effective cost is 8.3238%.

If the person reimburses after 10 years, the outstanding balance is £614,073, on which he will also pay a 3% charge. The IRR is 8.6442%. If repayment happens after 5 years, the balance is £665,489. With an additional 3% charge, this gives an IRR/effective cost of 9.2278%.

e. Is the pattern in your findings expected? And why do banks charge discount points? (5 points)

The pattern is expected. Banks charge discount points to insure against prepayment (ex-post) and attract immobile workers who are unlikely to prepay (avoid adverse selection, ex-ante). These issues were discussed in Lecture 5.

f. You are now considering a Graduated Payment Mortgage (GPM) with the following characteristics on the £1,000,000 purchase. The LTV is still at 70%, the expected duration is 30 years and the annual interest is 8%. However, there is a graduation rate of 5% per annum which applies to years 2, 3, 4, 5 and 6. From the beginning of year 6 till the end of the mortgage (in year 30), the payments then remain constant. Can you determine the value of the payments at the beginning of years 1, 2, 3, 4, 5 and 6? (25 points)

A similar mortgage structure was discussed during Lecture 5. The solution can be found considering that the PV of the loan is the presented discounted value of all future payments which increase at predetermined steps. The corresponding 'bundled' cash flow is as follows:

$$700{,}000 = PMT1 \times \sum_{t=1}^{12} \frac{1}{\left(1+\frac{8\%}{12}\right)^{t}}$$

$$PMT1(1+5\%) \times \sum_{t=1}^{12} \frac{1}{\left(1+\frac{8\%}{12}\right)^{t}} \times \frac{1}{\left(1+\frac{8\%}{12}\right)^{12}}$$

$$+PMT1(1+5\%)^2 \times \sum_{t=1}^{12} \frac{1}{\left(1+\frac{8\%}{12}\right)^{t}} \times \frac{1}{\left(1+\frac{8\%}{12}\right)^{24}}$$

$$+PMT1(1+5\%)^3 \times \sum_{t=1}^{12} \frac{1}{\left(1+\frac{8\%}{12}\right)^{t}} \times \frac{1}{\left(1+\frac{8\%}{12}\right)^{36}}$$

$$+PMT1(1+5\%)^4 \times \sum_{t=1}^{12} \frac{1}{\left(1+\frac{8\%}{12}\right)^{t}} \times \frac{1}{\left(1+\frac{8\%}{12}\right)^{48}}$$

$$+PMT1(1+5\%)^5 \times \sum_{t=1}^{300} \frac{1}{\left(1+\frac{8\%}{12}\right)^{t}} \times \frac{1}{\left(1+\frac{8\%}{12}\right)^{60}}$$

This can be rewritten as:

$$700{,}000 = PMT1 \times 11.4958 + PMT1 \times 1.05 \times 11.4958 \times 0.9234 + PMT1 \times 1.1025 \times 11.4958 \times 0.8526 + PMT1 \times 1.1576 \times 11.4958 \times 0.7873 + PMT1 \times 1.2155 \times 11.4958 \times 0.7269 + PMT1 \times 1.2765 \times 129.56 \times 0.6712 = 165.1239PMT1$$

Which means the first payment will be £4239.24; the second one will be £4,451.2; the third will be £4673.77; the fourth will be £4907.45; the fifth will be £5152.83; and all the others will be £5410.46.

g. What do you notice in the time schedule of the payments of the GPM relative to the payments of the payments of the equivalent CPM (discussed in point d)? Which mortgage does the bank prefer? Substantiate your answer to this second point by comparing the amount of outstanding balance after one year for the GPM and for the CPM, and by commenting on your finding. (20 points)

These issues were extensively discussed in Lecture 5. The outstanding amount of the balance for the CPM can be found using the amortization worksheet on the calculator (£694,152.45). The outstanding balance for the GPM has to be calculated using the FV function on the calculator considering that the mortgage for £700,000 is paying only £4239.24 for the first twelve months despite interest being charged at 8%. So the outstanding balance goes up. This amounts to £705,321.4. The bank prefers the CPM.

Question 3: You are thinking about investing in a small building with a total rentable space of 3500 square metres and a retail destination. You are planning to keep the asset for 5 years. You make the following forecasts/assumptions. The prevailing price per square metre per year is £200. The first tenant you expect to have will be a small supermarket occupying 2000 square metres of your space. This tenant will be given 1.5 years free of rent. When this space is occupied by the supermarket it generates an additional £20 per square metre per year in advertising; if the space is not occupied, this income is totally disrupted (i.e. the value is zero). The second tenant will be a small shop occupying 500 metres of your space from the first year onwards. No concessions or incentives will be given to this tenant. The third tenant is a clothing shop that will occupy (from the first year onwards) the remaining space and be charged full rent right from the beginning. From year 3, you will also impose an overage charge of 2% of its revenues on this tenant. These are expected to be at £200,000 from year 2 onwards.

a- Put together the operation and reversion pro-forma for this investment considering that your expected exit yield is 12%. (20 points)

These issues were discussed in several lectures. The exit yield is the same concept as the going out cap rate. The pro-forma for the operations looks as follows:

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Tenant 1	0	50% of 2000 m2 × £200= £200,000	2000 m2 × £200 = £400,000	2000 m2 × £200 = £400,000	2000 m2 × £200 = £400,000	2000 m2 × £200 = £400,000
Tenant 2	500m2 × £200 = £100,000	500m2 × £200 = £100,000	500m2 × £200 = £100,000	500m2 × £200 = £100,000	500m2 × £200 = £100,000	500m2 × £200 = £100,000
Tenant 3	1000m2 × £200 = £200,000	1000m2 × £200 = £200,000	1000m2 × £200 = £200,000	1000m2 × £200 = £200,000	1000m2 × £200 = £200,000	1000m2 × £200 = £200,000
Add. income	2000 m2 × £20 = £40,000	2000 m2 × £20 = £40,000	2000 m2 × £20 = £40,000	2000 m2 × £20 = £40,000	2000 m2 × £20 = £40,000	2000 m2 × £20 = £40,000
Overage	0	0	2% of £200,000 = £4,000	2% of £200,000 = £4,000	2% of £200,000 = £4,000	2% of £200,000 = £4,000
Total	£340,000	£540,000	£744,000	£744,000	£744,000	£744,000

With an exit yield of 12%, the sale price is £6.2mio.

b- If you are asked to pay £5.8million to purchase this building, what would be your Internal Rate of Return (IRR)? (5 points)

The IRR of this investment for a price of £5.8mio is 11.44%.

c- Consider the following alternative scenario. The space you expect rent to the supermarket remains vacant for the first two years. In year 3, you finally sign a tenant for this part of your building, but still need to provide 1.5 years free of rent. The absence of the supermarket completely disrupts your other sources of income from advertising, forces you to give a 20% discount to the second tenant (the small shop) while the main space is vacant, and reduces the revenues of the clothing shop which are expected to fall to £100,000 between years 1 and 4, but and recover to £200,000 in year 5. Build the cash flow for this scenario. If you have the same exit yield and purchase price, what are the sale price and the IRR of this investment? What do you notice compared to the findings at points a and b? (20 points)

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Tenant 1	0	0	0	50% of 2000 m2 × £200= £200,000	2000 m2 × £200 = £400,000	2000 m2 × £200 = £400,000
Tenant 2	500m2 × £200 × (1-20%) = £80,000	500m2 × £200 × (1-20%) = £80,000	500m2 × £200 = £100,000	500m2 × £200 = £100,000	500m2 × £200 = £100,000	500m2 × £200 = £100,000
Tenant 3	1000m2 × £200 = £200,000	1000m2 × £200 = £200,000	1000m2 × £200 = £200,000	1000m2 × £200 = £200,000	1000m2 × £200 = £200,000	1000m2 × £200 = £200,000
Add. income	0	0	2000 m2 × £20 = £40,000	2000 m2 × £20 = £40,000	2000 m2 × £20 = £40,000	2000 m2 × £20 = £40,000
Overage	0	0	2% of £100,000 = £2,000	2% of £100,000 = £2,000	2% of £200,000 = £4,000	2% of £200,000 = £4,000
Total	£280,000	£280,000	£342,000	£542,000	£744,000	£744,000

The sale price on this investment remains 6.2mio because the NOI in year 6 has not changed. But all the rest has been affected, so the IRR is lower at 8.3878%.

d- If the two alternative scenarios happen with 50% probability, what is the expected IRR? And what are the variance, standard deviation and coefficient of variation on the return on this investment? (5 points)

These issues were discussed in Lecture 3 and 4, and similar calculations were carried out in the classes. The answers are as follows: E(IRR)=9.9139; Var(IRR)=2.239; Std.Dev.=1.526; CoV=0.1539.

e- Assume now you leverage your investment. You take an interest-only loan with a loan-to-value (LTV) ratio of 50% and charging an annual interest of 10% (assume annual compounding). What is the IRR of the leveraged investment in the optimistic scenario used at point a? What is the IRR of the leveraged investment for the pessimistic scenario described at point c? (15 points)

The loan structure is very simply. The amount borrowed is 50% of £5.8mio, so £2.9mio. The annual interest payments at a 10% rate are £290,000. The students simply need to remember to subtract £290,000 mio from the NOI of every year, consider that the equity invested is only £2.9mio and that when they sell the net price will be £6.2mio – £2.9mio = £3.3mio. They should also note that for the first two periods their cash flow will be negative. With these assumptions, the IRR in the positive scenario is 12.74% and in the negative scenario is 6.9488%.

f- What are the expected IRR, standard deviation and coefficient of variation of the leveraged investment? Comments on the effects of leverage on the returns and risks. (10 points)

The E(IRR) is now 9.844%, which is not very different from the one obtained at point d. However, the variation in the investment returns is much higher. The std. dev. is now 2.8956 and the CoV is 0.2941. This is expected: leverage amplifies risks and returns. Since in one case here leverage has a negative effect (rp<rd → 8.3878%<10%), the effect of leverage is predominantly to magnify risks.

g- Using the Weighted Average Cost of Capital (WACC) formula, show what the implied returns on equity are in the optimistic and pessimistic scenarios. Are these numbers the same as the IRR you obtained at points b and c? Why/why not? (10 points)

The WACC formula would give the following numbers:

Optimistic: 11.44% + (11.44% - 10%) × D/E =11.44% + (1.44%) × 1 = 12.88%

Pessimistic: 8.3878% + *(8.3878% - 10%) × D/E = 6.7756%

The results are close but not the same as the IRR. This is because WACC is an approximation.

h- Focus now again on the pessimistic scenario and consider that every time your operating cash flow is below the money required to pay interests on the loan, you have to 'draw down' your equity to make sure the negative result effectively becomes a zero. Note that this operation will essentially leave you will less equity once you have sold the asset and repaid the loan at the end of in year 5 (since you have used part of it to eliminate negative operation cash flow results). What is your IRR in this case? (15 points)

Given how the question is formulated, the only two ingredients that change relative to the standard pessimistic scenario are the following.

The cash flows of periods 1 and 2 now become 0 (and not negative £10,000). The final sale now sees a price of 6.2mio to reimburse £2.9mio loan but the equity left is less than £2.9mio since £20,000 (it is instead £2.88mio) have been drawn down to compensate for two negative cash flows. The IRR of this investment is 6.9773%.

Insightful students should comment that this is larger than the previous IRR since the loss in the equity is discounted back from year 5, but I avoid losses closer to time 0 – in year 1 and 2 – which have a larger impact on the IRR.

Note that equity draw down could be modelled differently. For example, students could have modelled the possibility that at time zero the investor is already anticipating losses in future periods and money is requested in advance. The discounting here would be different.

In assessing the answer to this question, the marker would have looked for clear evidence that the students can think about the problem at hand.